

Assignment 5 – BMI Program Test Plan

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Course: CIS 206 – Applied Programming

Program: BMI_Activity4.py

Testing Framework: Python unit test

1. Purpose

The purpose of this test plan is to test the BMI program and make sure that it correctly handles valid input, invalid input, BMI calculations, and the option to quit the program.

2. Testing Technology

I chose Python's **unit test** testing framework. It is included with Python and allows me to create automated test cases and check whether the program produces the expected results.

3. Test Cases

Test Case 1 – Valid BMI Calculation

Input:

Weight: 70 kg

Height: 1.75 meters

Expected Result:

The program should display a BMI of **22.86**.

Result: Passed

Test Case 2 – Negative Weight

Input:

Weight: -5 kg

Expected Result:

The program should display:

“Invalid input. Please try again.”

Result: Passed

Test Case 3 – Non-Numeric Weight

Input:

Weight: ABC

Expected Result:

The program should display:

“Invalid input. Please enter a number.”

Result: Passed

Test Case 4 – Quit Option

Input:

Q

Expected Result:

The program should end and display:

“Program ended.”

Result: Passed

4. Test Execution

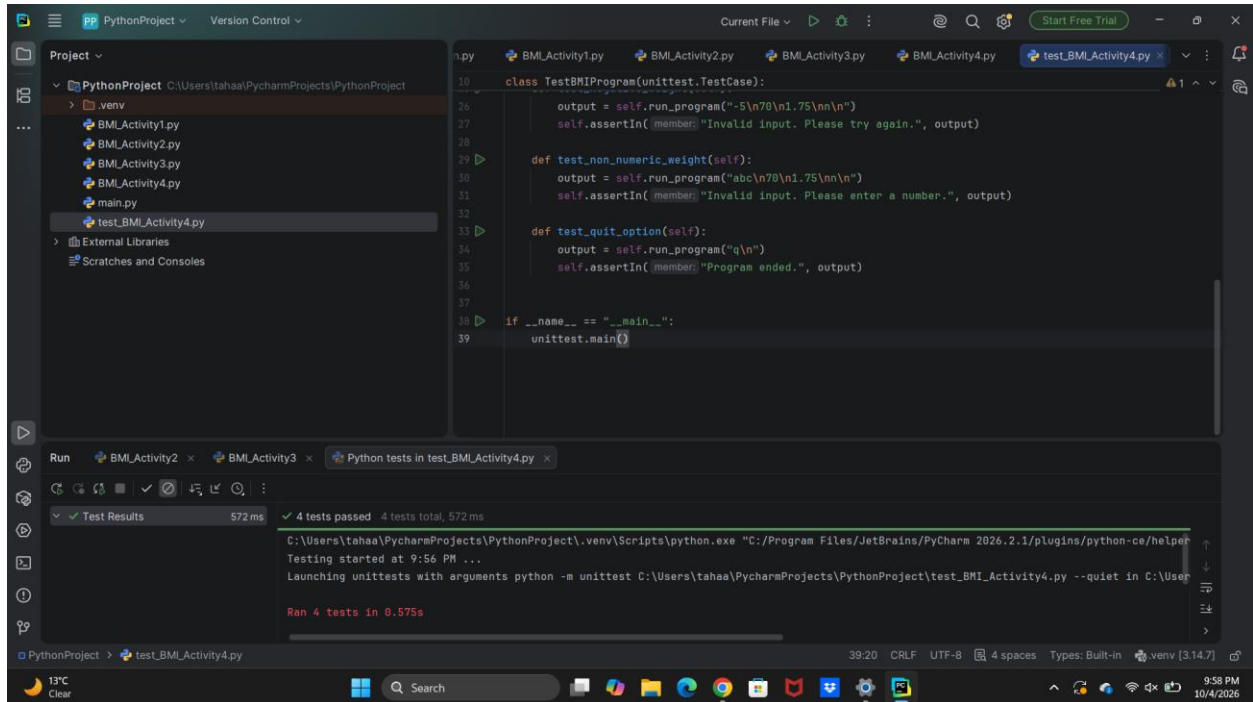
The test cases were implemented using Python's unit test framework. The tests were executed in PyCharm.

The test execution resulted in:

4 tests passed

Ran 4 tests successfully.

5. Test Results



6. Conclusion

All four test cases passed successfully. The testing confirms that the BMI program can handle valid input, invalid input, negative weight, and the quit option as expected.